

Result

(20,0)

Q1: Calculate the Euclidean distance between the user and the satellite

Distance = 1200010.314331 m

Q2: Compute the path loss

Pathloss (dB) = -157.553308

No beamforming Pathloss (dB) = -181.635693

Q3: Compare Rx power w/ and w/o beamforming

Q4: Evaluate the SNR & data rate

下面是有做 beamforming 的數據

```
RxPower    = -56.8133 dBm
SNR         = 53.1867 dB
DataRate   = 3.53365e+07 bps
```

下面是沒有做 beamforming 的數據

```
RxPower    = -80.8957 dBm
SNR         = 29.1043 dB
DataRate    = 1.934e+07 bps
```

Q5: Estimate the end-to-end delay

沒有做 beamforming 下，平均 end to end delay 是 0.401509s

```
Packet 0 end-to-end delay is 0.0900081 s
Packet 1 end-to-end delay is 0.164009 s
Packet 537 end-to-end delay is 1.26601 s
Packet 1026 end-to-end delay is 0.0860088 s
Average end-to-end delay is 0.401509 s
```

有做 beamforming，平均 end to end delay 是 0.00808541s

```
Packet 0 end-to-end delay is 0.00800872 s
Packet 1 end-to-end delay is 0.00800931 s
Packet 537 end-to-end delay is 0.00829733 s
Packet 1026 end-to-end delay is 0.00802628 s
Average end-to-end delay is 0.00808541 s
```

(6.06692, 73.0213)

Distance = 1200012.502119 m

Pathloss (dB) = -157.830023

No beamforming Pathloss (dB) = -181.635744

下面是有做 beamforming 的數據

```
RxPower    = -57.09 dBm
SNR         = 52.91 dB
DataRate    = 3.51526e+07 bps
```

下面是沒有做 beamforming 的數據

```
RxPower    = -80.8957 dBm
SNR         = 29.1043 dB
DataRate    = 1.934e+07 bps
```

沒有做 beamforming 下，平均 end to end delay 是 0.401661s

```
Packet 0 end-to-end delay is 0.0901749 s
Packet 1 end-to-end delay is 0.164154 s
Packet 537 end-to-end delay is 1.26616 s
Packet 1026 end-to-end delay is 0.0861542 s
Average end-to-end delay is 0.401661 s
```

有做 beamforming，平均 end to end delay 是 0.00825102s

```
Packet 0 end-to-end delay is 0.00817493 s
Packet 1 end-to-end delay is 0.00817498 s
Packet 537 end-to-end delay is 0.00846154 s
Packet 1026 end-to-end delay is 0.00819261 s
Average end-to-end delay is 0.00825102 s
```

(-16.0634, 142.29)

Distance = 1200004.114826 m

Pathloss (dB) = -157.812488

No beamforming Pathloss (dB) = -181.635693

下面是有做 beamforming 的數據

```
RxPower    = -57.0725 dBm
SNR        = 52.9275 dB
DataRate   = 3.51643e+07 bps
```

下面是沒有做 beamforming 的數據

```
RxPower    = -80.8957 dBm
SNR        = 29.1043 dB
DataRate   = 1.934e+07 bps
```

沒有做 beamforming 下，平均 end to end delay 是 0.401509s

```
Packet 0 end-to-end delay is 0.0900081 s
Packet 1 end-to-end delay is 0.164009 s
Packet 537 end-to-end delay is 1.26601 s
Packet 1026 end-to-end delay is 0.0860088 s
Average end-to-end delay is 0.401509 s
```

有做 beamforming，平均 end to end delay 是 0.00808578s

```
Packet 0 end-to-end delay is 0.00800872 s
Packet 1 end-to-end delay is 0.00800931 s
Packet 537 end-to-end delay is 0.00829873 s
Packet 1026 end-to-end delay is 0.00802637 s
Average end-to-end delay is 0.00808578 s
```

可以發現有 beamforming 的 delay 約為沒有的 1/50 倍，但 data rate 的速度應該差了約 5000 倍，實際上總 delay 並沒有減少那麼多倍。我推測是因為傳輸所花的時間已經非常小了，加快速度不會明顯改變時間，剩下占比最大的時間可能是 Propagation delay 等 delay 造成的，而我們只減少了傳輸的 delay，其他的 delay 我們沒有減少，才會有這樣的結果。